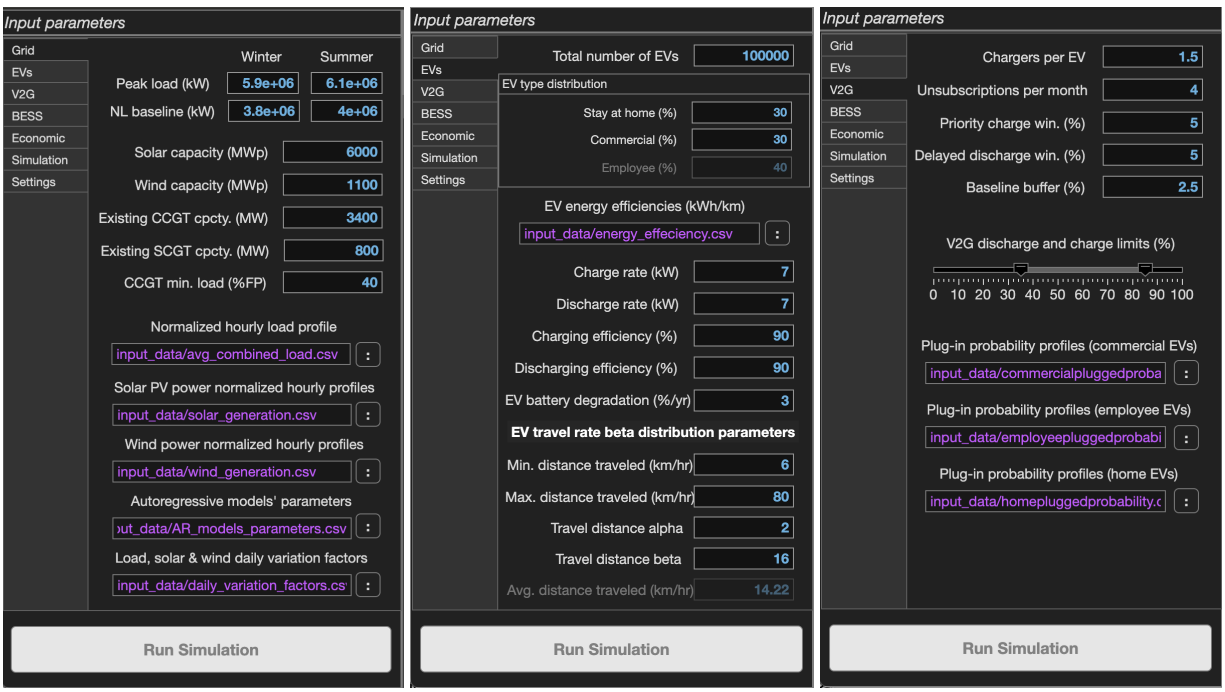
GSV2GS input format guide

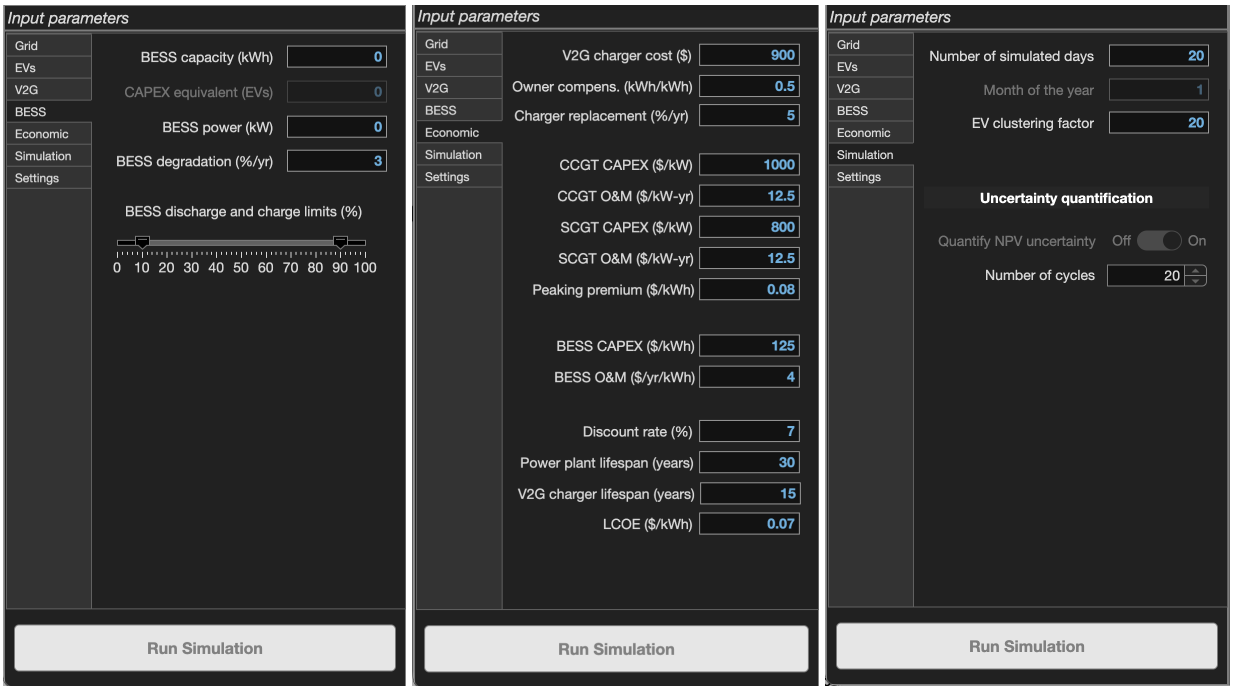
This file describes the proper formatting for the various inputs which are required to run simulations using this grid-scale V2G simulator. More information about the inputs and the related assumptions can be found in the paper. Sample input files are provided in the repo which could be used as a starting point to a modified simulation

> *Required .csv files must be located in a folder named "input_data" located within the directory of the source-file (Matlab) / application (macOS) or .exe (windows).*

Figure 1 shows the six input panels within the software, and Table 1 to Table 6 describe each of the inputs and explain the required formating.



(a) (b) (c)



(d) (e) (f)

Figure 1: Screenshots of GSV2GS input panels. Table 1 to Table 6 provide elaborations for panels (a) to (f), respectively.

Table 1: Description and proper formatting of input "grid parameters".

Input	Unit	Description	Format
Peak-load	kW	The maximum anticipated demand-load	Two values for summer and winter seasons must be provided through input text fields
NL baseline	kW	The net-load baseline, which is a central aggregator parameter which defines the priority function of the V2G fleet, i.e. load-following vs peak shaving	Two values for summer and winter seasons must be provided through input text fields
Solar capacity	MWp	Total installed solar PV nameplate capacity	Single value provided through an input text field
Wind capacity	MWp	Total installed wind farm nameplate capacity	Single value provided through an input text field
Existing CCGT cpcty.	MW	Total pre-existing load-following capacity	Single value provided through an input text field
Existing SCGT cpcty.	MW	Total pre-existing peaking capacity	Single value provided through an input text field

CCGT min. load	%FP	The minimum power level (fraction of full power) the load following capacity can operate at.	Single value provided through an input text field
Normalized hourly load profiles	Unitless	Normalized hourly profiles representing the shape of the average total demand-load over the power grid	The data must be provided through ***.csv*** file which consists of five columns: **Column 1:** time series column (0 -> 24 hours) showing the time values corresponding to columns 2 to 5. **Column 2:** normalized demand-load value for **weekdays in winter season** **Column 3:** normalized demand-load value for **weekends in winter season** **Column 4:** normalized demand-load value for **weekdays in summer season** **Column 5:** normalized demand-load value for **weekends in summer season**
Solar PV power normalized generation profiles	kW/MWp	Normalized hourly profiles representing the shape of the average power production through solar PV	The data must be provided through ***.csv*** file consisting of 13 rows: **Row 1:** time series row (0 -> 24 hours) showing the time values corresponding to rows 2 to 13. **Rows 2 to 13:** normalized PV power generation profiles over the 12 months of the year, from January till December. The values are in kW per MWp nameplate capacity.
Wind power normalized generation profiles	kW/MWp	Normalized hourly profiles representing the shape of the average power production through installed wind farms	The data must be provided through ***.csv*** file consisting of 13 rows: **Row 1:** time series row (0 -> 24 hours) showing the time values corresponding to rows 2 to 13. **Rows 2 to 13:** normalized wind power generation profiles over the 12 months of the year, from January till December. The values are in kW per MWp nameplate capacity.
Autoregressive model's parameters	Unitless	First-order autoregressive models' parameters for the simulation of *stochastic* hourly variability in	The data must be provided through ***.csv*** file consisting of 3 rows and 6 columns: **Row 1:** lists the autocorrelation Coefficients (ρ) **Row 2:** lists the "innovation noise factors" (σ)

		demand-load, solar, and wind generation	**Row 3:** lists fractions which dictate the range over which the values vary. Example: for a fraction of 0.15, the generated variability factors would range from 0.85 to 1.15. **Columns 1 to 3:** list the parameters for demand-load, solar and wind AR(1) models, respectively, during winter season **Columns 4 to 6:** list the parameters for demand-load, solar and wind AR(1) models, respectively, during summer season
Load solar and wind daily variation factors	Unitless	Lists of factors which enable artificially increasing or decreasing the average load or generation over a whole day in order to simulate various weather-related phenomena	The data must be provided through ***.csv*** file consisting of 3 columns, for demand-load, solar PV and wind, in the listed order. The number of rows must be at least equal to the number of days to be simulated as prescribed by the user

Table 2: Description and proper formatting of input "EV parameters".

Input	Unit	Description	Format
Total number of EVs	EVs	Total number of subscribed EVs	Single value provided through an input text field
EV type distribution	Percentage	Fractions of three different EV types	Two percentage values provided through input text fields. The third percentage is calculated automatically by the software to complete a 100%.
EV energy efficiencies	kWh/km	Amount of energy consumed by EV per kilometer travelled.	The data must be provided through ***.csv*** file consisting of 2 columns for winter and summer seasons, respectively. The data must consist of three rows, corresponding to 40, 60 and 80 kWh battery EVs.
Charge and discharge rate	kW	The power rating of the V2G charger in charging and discharging directions	Two values provided through input text fields

Charging and discharging efficiencies	Percentage	The efficiency of the charging and discharging processes	Two values provided through input text fields
EV battery degradation rate	Percentage	The annual loss in battery capacity as a percentage of its original capacity.	Single value provided through an input text field
EV travel rate beta distribution parameters	Assorted	Parameters of the beta distribution modelling the travel rate probability distribution	Four values provided through input text fields: (1) minimum travel speed (km/hr), (2) max travel speed (km/hr), (3) alpha parameter of the beta distribution and (4) beta parameter of the beta distribution. The software automatically calculates the corresponding average distance travelled based on the prescribed parameters.

Table 3: Description and proper formatting of input "V2G parameters".

Input	**Unit**	**Description**	**Format**
Chargers per EV	Chargers	Number of V2G chargers installed per subscribed EV	Single value provided through an input text field
"Unsubscriptions" per month	Count	The number of times a subscribed EV may opt to override the central aggregator – for a duration of 24 hours	Single value provided through an input text field
Priority charge and delayed discharge windows	Percentage	Percentages define the window around the net-load baseline within which priority charge and delayed discharge are executed. The percentage is multiplied by the net-load baseline value to evaluate the window size in kW	Two values provided through input text fields
Baseline buffer	Percentage	Percentage value defines the window around the net-load	Single value provided through an input text field

		baseline within which EVs are neither discharged nor charged. The percentage is multiplied by the net-load baseline value to evaluate the window size in kW	
V2G discharge and charge limits	SOC percentage	The lower and upper SOC values which define the V2G SOC window	Two values selected using a slider bar.
Plug-in probability profiles (three EV types)	Probability	Hourly profiles defining the probability an EV is plugged-in to an EV charger at a particular time during the day	The data must be provided through three ***.csv*** files (one for each EV type) Each file consists of four columns, representing profiles for: (1) summer weekday, (2) summer weekend, (3) winter weekday and (4) winter weekend, in the mentioned order. The data must consist of 25 rows describing the probability at each hour of the day (first and last row are copies)

Table 4: Description and proper formatting of input "BESS parameters".

Input	**Unit**	**Description**	**Format**
BESS capacity	kWh	Total stationary BESS capacity.	Single value provided through an input text field. To assist with comparisons to V2G economic performance, the software converts the prescribed BESS capacity into a number of EVs which have the equivalent CAPEX investment
BESS power	kW	The maximum power of the BESS	Single value provided through an input text field
BESS degradation	Percentage	Annual degradation rate of BESS installation	Single value provided through an input text field

		in percentage of the initial capacity	
BESS discharge and charge limits	SOC percentage	The lower and upper SOC values which define the BESS SOC window	Two values selected using a slider bar.

Table 5: Description and proper formatting of input "economic parameters".

Input	Unit	Description	Format
V2G charger cost	$	V2G charger unit cost	Single value provided through an input text field
Owner compens.	kWh per kWh	The amount of free kWh rewarded as a compensation to the EV owner per kWh discharged to the grid	Single value provided through an input text field
Charger replacement	Percentage	Annual rate at which V2G chargers might fail and would require replacement. This rate may also be considered to include maintenance costs of V2G chargers	Single value provided through an input text field
CCGT/SCGT CAPEX	$/kW	The capital expenditure cost associated with building load-following/peaking power plants, per kW of rated power	Values provided through input text fields
CCGT/SCGT O&M	$/kW/year	The annual operation and maintenance costs associated with operating load-following/peaking power plants, excluding fuel costs, per kW of installed capacity	Values provided through input text fields
Peaking premium	$/kWh	The cost difference between generating electricity using load-following and peaking power plants	Single value provided through an input text field

BESS CAPEX	$/kWh	The capital expenditure cost associated with building BESS, per kWh of capacity	Single value provided through an input text field
BESS O&M	$/kWh	The annual operation and maintenance costs associated with operating BESS, per kWh of capacity	Single value provided through an input text field
Discount rate	Percentage	Annual interest rate to account for time value of money	Single value provided through an input text field
Power-plant/V2G-charger lifespan	Years	Life expectancy of installed conventional-power plant infrastructure/V2G chargers after which the equipment will have to be decommissioned	Single value provided through an input text field
LCOE	$/kWh	Average levelized cost of generating electrical power per kWh	Single value provided through an input text field

Table 6: Description and proper formatting of input "simulation parameters".

Input	**Unit**	**Description**	**Format**
Simulated days	Days	Total number of days to be simulated (per cycle) for which data has been provided	Single value provided through an input text field
EV clustering factor	Number	A clustering factor which combines multiple EVs into lumps in order to accelerate the simulation	Single value provided through an input text field
Number of cycles	Number	The number of times the simulation entire duration is repeated in order to achieve statistical convergence	Single value provided through an input text field